Prime Numbers

The Sieve of Eratosthenes

Session 2: The Magic Number Filter

Session Introduction

In our last session, we learned about the 'Builders' (Factors) and the 'Families' (Multiples). Today, we are going to use a magic filter called a 'Sieve' to find special numbers that refuse to be broken down—the **Prime Numbers!**

Learning Journey

Today, you will become a "Math Detective" as we explore these goals:

- **Discover Soloists:** Understand what makes a Prime Number unique.
- **Identify Composites:** Learn about numbers that have many factors.
- **Use the Magic Filter:** Practice the step-by-step Sieve of Eratosthenes.
- **Crack the Case of Number 1:** Solve the mystery of why 1 is so different!

DID YOU KNOW?

Eratosthenes was a brilliant thinker who lived over 2,000 years ago! He invented this 'Sieve' method to catch prime numbers just like a fisherman catches fish in a net.

The Prime Soloists

Some numbers are like solo performers on a stage. They don't like to be divided into equal groups of 2, 3, or any other number.

> **Definition:** A **Prime Number** is a whole number greater than 1 that has exactly two factors: 1 and itself.

The Kabaddi Analogy

Imagine 7 children playing **Kabaddi** in the school playground.

- Can they make 2 equal teams? No (3 and 3, but 1 is left).
- Can they make 3 equal teams? No (2, 2, and 2, but 1 is left).
- The ONLY way they can stand is as 1 line of 7 children or 7 separate individuals.

Because 7 can only be divided by 1 and 7, it is a ***Prime Number***!

Composite Numbers: The Team Players

While some numbers are soloists, others love to form teams! These are called **Composite Numbers**.

> **Definition:** A **Composite Number** is a number that has more than two factors.

Building with 6

Think of 6 Chikki pieces. We can arrange them in many ways:

- 1 row of 6 ***(1 × 6)***
- 2 rows of 3 ***(2 × 3)***
- 3 rows of 2 ***(3 × 2)***
- 6 rows of 1 ***(6 × 1)***

Factors of 6 are **1, 2, 3, and 6**. Since there are more than two factors, 6 is a ***Composite Number***!

The Magic Filter: Sieve of Eratosthenes

Imagine your mother in the kitchen. When she makes tea, she uses a **'Channi'** (sieve) to filter out the tea leaves from the tea.

[Visual: Illustration of a Sieve filtering numbers]

The Sieve of Eratosthenes works the same way. It "filters out" all the composite numbers, and only the "pure" prime numbers remain in our grid!

HOW IT WORKS

We start with a grid of numbers. We circle a number (a Prime) and then "cross out" all of its multiples (the Composites). We repeat this until only the Primes are left!

Let's Start Filtering!

Follow these steps on your number grid (1 to 50):

Step 1: The Mystery of Number 1

Cross out **1**. It is not prime because it doesn't have two different factors (it only has one: 1!).

Step 2: The Only Even Prime

Circle **2** (It's prime!). Now, cross out all multiples of 2 (4, 6, 8, 10, 12...). These are all composite because they can be divided by 2!

Step 3: Moving to Three

The next available number is **3**. Circle it! Now, cross out all multiples of 3 (6, 9, 12, 15...). Some might already be crossed out—that's okay!

Continuing the Sieve

Step 4: Fantastic Five

Circle **5**. Cross out its multiples (10, 15, 20, 25...).

Step 5: Lucky Seven

Circle **7**. Cross out its multiples (14, 21, 28, 35, 42, 49).

The Result

After filtering by 2, 3, 5, and 7, all the numbers that haven't been crossed out in your 1-50 grid are **PRIME NUMBERS!**

Your Prime Treasure Map (1-50)

Here are the "Soloists" we caught in our sieve:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

MATH DETECTIVE PRACTICE

Look at the number **13**. Can you find any way to divide 13 Rangoli dots into equal groups? Try 2s, 3s, 4s...
No? That's because 13 is a Prime Soloist!

The Legend of Number 1

In the world of numbers, **1** is a bit of a rebel. It doesn't follow the rules of Prime or Composite.

> To be **Prime**, you need 2 factors (1 and self).
>
> To be **Composite**, you need MORE than 2 factors.
>
> **Number 1** only has 1 factor (itself!).

So, we say: **"1 is neither Prime nor Composite!"** It is unique, just like you!

Session Recap

You've done a fantastic job today, Detective!

- **Prime Numbers:** Only 2 factors (e.g., 2, 3, 5, 7).
- **Composite Numbers:** More than 2 factors (e.g., 4, 6, 8, 9).
- **The Sieve:** A magic way to filter out composites.
- **Special Rule:** 1 is unique!

THE PLAYGROUND CHALLENGE

Next time you are at the park with 11 friends (total 12), try to form different size groups. Then, try it if there are only 11 of you. Which one is easier? Why?